

MEDASSISTAI

AI-Assisted Healthcare Platform

MILESTONE 4 • FINAL DEPLOYMENT & SYSTEM DOCUMENTATION

Project	MedAssistAI
Team	Team 2
Hosted Website	https://medassistai-frontend.vercel.app
Milestone Scope	System theory, architecture, deployment and verification

Project Aim

MedAssistAI is an AI-assisted healthcare platform that connects patient symptom analysis, disease prediction, severity and risk information, doctor consultation, appointment management, reporting, profile updates, and post-consultation medication guidance within one workflow.

This milestone focuses on the theoretical system design and the deployed architecture used to make the application accessible for demonstration and evaluation.

Production Website

medassistai-frontend.vercel.app

1. Project Overview and Functional Scope

MedAssistAI provides a structured digital healthcare experience for patients while supporting connected doctor and administrator workflows. The patient can create an account, sign in, analyze symptoms, receive an AI-assisted prediction, review severity and risk information, download a report, request a consultation, manage appointments, update profile information, and view medication guidance after doctor confirmation.

Core Functional Areas

Area	Purpose
Patient Access	Registration, login, profile information and patient dashboard.
Symptom Analysis	Search/select multiple symptoms and submit them for analysis.
Disease Prediction	Process symptoms through the trained prediction workflow and return the relevant result.
Severity & Risk	Present severity score/level and risk information.
Report	Download the generated analysis report.
Appointments	Select a doctor, specialization, date/time and consultation details, then manage appointments.
Doctor Connection	Doctor confirmation connects the appointment workflow with follow-up medication/suggestion information.
Profile	Patient profile details can be updated when required.

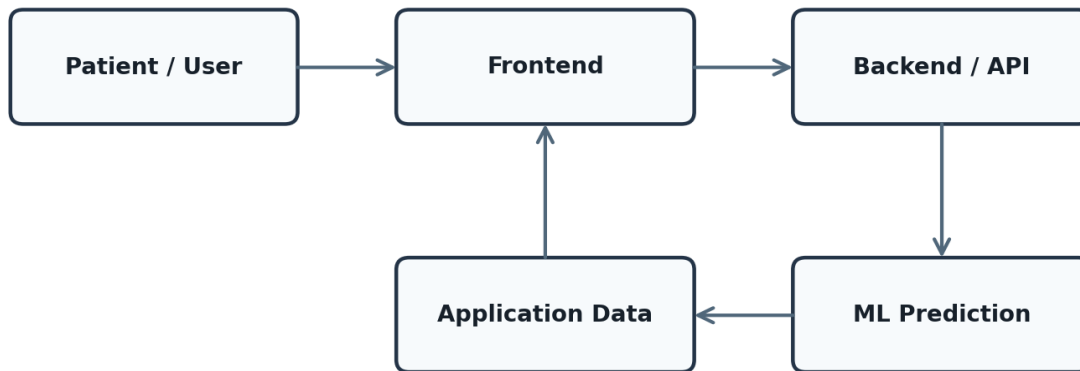
Role Model

Patients can register themselves. Admin and Doctor access is restricted to authorized accounts defined by the project design. An Admin can create additional Doctor accounts whenever required.

2. System Architecture and Patient Workflow

The system is organized as a connected web application. The user interface communicates with the application service, while the prediction workflow provides AI-assisted analysis. The deployed components are separated so the frontend, service layer, and model layer can be maintained independently.

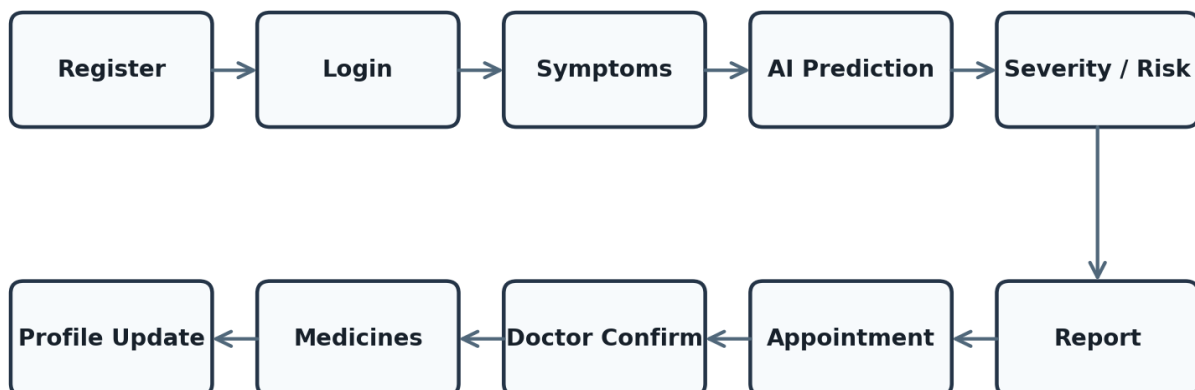
MedAssistAI High-Level System Architecture



Patient Journey

The patient workflow begins with account registration and login and continues through symptom analysis, prediction, severity/risk review, reporting, appointment booking, doctor confirmation, medication guidance, and profile management.

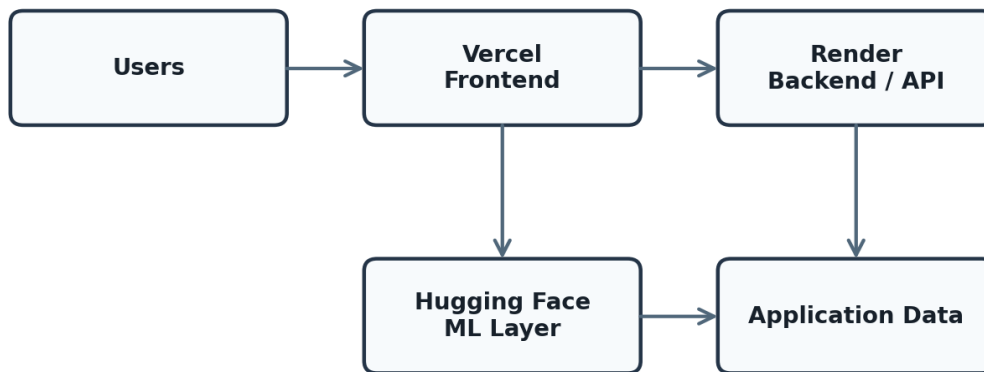
Patient-Centric Healthcare Workflow



3. Deployment Architecture

For Milestone 4, the application has been deployed as a hosted system so the complete workflow can be inspected through a live URL. The frontend is deployed through Vercel. The application/backend service is deployed through Render, and the machine-learning model component is included in the Hugging Face deployment/hosting setup used for the project.

Deployment and Service Integration



Deployment Components

Component	Deployment / Role
Frontend	Vercel • hosts the production web interface and public website.
Backend / API	Render • hosts the application service used by the deployed system.
ML Layer	Hugging Face • used in the project's model deployment/hosting setup.
Public Access	The live application is accessible through the Vercel production domain.

Production URL

<https://medassistai-frontend.vercel.app>

4. Deployment Verification and Milestone Status

The deployed application was checked as a complete user-facing system. Verification focused on accessibility of the hosted application, major role-based workflows, responsive behavior, and continuity between patient actions and doctor-side processing.

Verification Checklist

Check	Status
Production website accessible	Completed
Patient registration and login flow	Checked
Symptom analysis and disease prediction	Checked
Severity / risk information and report generation	Checked
Appointment creation and management	Checked
Doctor/Admin role access model	Checked
Responsive frontend behavior	Checked on mobile and desktop
Deployment structure documented	Completed

Milestone Outcome

Milestone 4 establishes MedAssistAI as a demonstrable hosted healthcare application rather than only a local development build. The production frontend is available through Vercel, the application service is deployed through Render, and the model layer is included in the Hugging Face deployment setup. The resulting architecture supports a clear demonstration path from patient access to AI-assisted analysis and onward to doctor consultation.

The current Vercel domain is sufficient for presentation and evaluation. A custom domain can be configured later as a separate deployment step without changing the core application workflow.

Project Website

<https://medassistai-frontend.vercel.app>

End of Milestone 4 Documentation